

TOH YU HENG



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<https://www.linkedin.com/in/yuheng-toh/>

<https://yhnothere.github.io/react-web-portfolio/>

EDUCATION

SINGAPORE INSTITUTE OF TECHNOLOGY

Computer Science in Real-Time Interactive Simulation

2023 - Current

SINGAPORE POLYTECHNIC

Diploma in Game Design and Development

2018 - 2021

LANGUAGES

C++, C, C#

Python

JavaScript

HTML, CSS

Kotlin

English, Chinese

PROFILE

Motivated Computer Science student with hands-on experience building custom game engines and real-time systems. Seeking a 1-year internship to apply strong systems and graphics fundamentals while contributing to production-level engineering projects.

WORK EXPERIENCE

INTERN

Vopak | Mar 2020 - Aug 2020

Developed an Augmented Reality (AR) application designed to assist workers in on-site learning and training.

- Designed and implemented a system that allowed workers and visitors to scan for verification, with data automatically sent and formatted in Excel for easier monitoring and tracking.
- Improved workflow efficiency by automating data collection processes and minimizing manual input errors.

PROJECTS

3D CUSTOM GAME ENGINE (CURRENT)

Developed a 3D game engine with emphasis on advanced physics implementation and real-time debugging tools.

- Integrated NVIDIA PhysX for rigid body dynamics, collision detection, and constraint solving.
- Implemented GPU-accelerated particle simulation using NVIDIA Flex.
- Developed real-time physics debug visualization (colliders, contact points, constraints).
- Designed a modular, user-friendly engine architecture to streamline gameplay feature development.

SOFTWARE

Visual Studio

Visual Studio Code

Unity

Unreal Engine

React, Node.js

Android Studio

Microsoft Excel

Blender

Git

2D CUSTOM GAME ENGINE

Designed and implemented a 2D game engine from scratch, leading development of the core physics system.

- Implemented rigid body dynamics with custom collision detection and response algorithms.
- Developed camera system and 2D lighting pipeline to enhance scene rendering and gameplay clarity.

3D PROCEDURAL CAVE GENERATION

Implemented procedural terrain generation using Binary Space Partitioning (BSP) and Cellular Automata to generate realistic 3D cave systems.

- Optimized generation performance and tuned Cellular Automata rules to produce natural, varied cave formations.
- Integrated Perlin Noise-based path generation into the cave system pipeline.

3D MULTIPLAYER GAME

Developed a 3D multiplayer "last-man-standing" party game using Unity and Photon Networking.

- Implemented player networking systems including synchronization, matchmaking, and in-game communication.
- Optimized network event handling and data transfer to ensure stable multi-client gameplay.
- Collaborated on integrating networked gameplay with movement systems, UI, particle effects, and core game flow.

BILL SPLITTER WEB APPLICATION

Developed a React-based bill splitting application that simplifies shared expenses through manual entry and automated receipt scanning.

- Implemented dynamic participant- and item-based expense splitting logic.
- Integrated OCR-based receipt scanning using Tesseract.js to extract item names and prices from images.
- Improved text recognition accuracy and handled edge cases such as uneven splits and incomplete data.
- Integrated real-time currency conversion using Google Finance data via Excel API to ensure up-to-date exchange rates.